

## Recall: Key Definitions

**Hamiltonian cycle.** A *Hamiltonian cycle* in a graph  $G$  is a cycle that contains *every* vertex of  $G$  (each vertex visited exactly once).

Unlike Eulerian tours (which cover every *edge*), Hamiltonian cycles cover every *vertex*. No efficient algorithm is known for deciding whether a graph has a Hamiltonian cycle.

**Line graph.** For a graph  $G = (V, E)$ , the *line graph*  $L(G)$  has vertex set  $E$  (the edges of  $G$ ), and two vertices  $e, e' \in E$  are adjacent in  $L(G)$  iff  $e \cap e' \neq \emptyset$  (i.e. the edges share an endpoint in  $G$ ).

**Eulerian graph.** A connected graph is *Eulerian* iff every vertex has even degree (Theorem 4.4.1).

## Practice: Textbook Exercise 4.4.8 — Line graphs, connectivity, and Hamiltonian cycles

For a graph  $G$ , let  $L(G)$  denote its line graph. Decide whether the following is true for every graph  $G$ :

- (a)  $G$  is connected if and only if  $L(G)$  is connected.
- (b)  $G$  is Eulerian if and only if  $L(G)$  has a Hamiltonian cycle.

### Remarks

**For part (a):** Think about what a path in  $G$  corresponds to in  $L(G)$ , and vice versa. What happens when  $G$  has an isolated vertex?

**For part (b):** An Eulerian tour in  $G$  visits every *edge*; edges of  $G$  are *vertices* of  $L(G)$ . What does the Eulerian tour become in  $L(G)$ ?

- (a) Prove that every graph  $G$  with  $n$  vertices and with all vertices having degree at least  $\frac{n}{2}$  has a Hamiltonian cycle.
- (b) Is it sufficient to assume degrees at least  $\lfloor n/2 \rfloor$  in part (a)?

 Remarks

**Strategy for (a).** Consider a *longest path*  $P = v_1 v_2 \cdots v_t$  in  $G$ .

- First show that  $t = n$  (the path visits every vertex).
- Then show that  $v_1$  and  $v_t$  have a common “rotation” index: if  $v_1$  is adjacent to  $v_k$ , then check whether  $v_t$  is adjacent to  $v_{k-1}$ . Use a **pigeonhole** / counting argument on the neighbors of  $v_1$  and  $v_t$  among  $\{v_1, \dots, v_t\}$ .

**For (b):** Try small cases. What happens when  $n$  is odd?